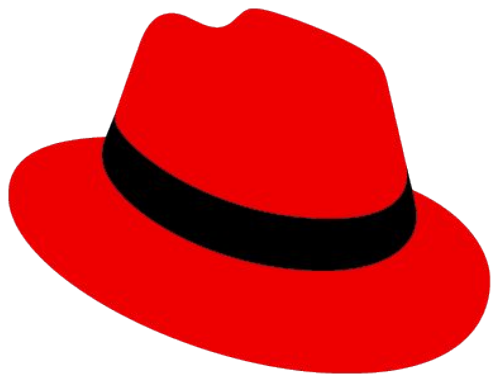




KubeArchive

**What it is,
and also what it isn't**

Topics

- Why KubeArchive?
- What does KubeArchive do?
- What KubeArchive is not!
- KubeArchive for Konflux



Why KubeArchive?

- Inspired by Tekton Results
 - Brainchild of Adam Kaplan
- Open source project
 - First commit in March of 2024
 - Team of five engineers
- Presented at Devconf
 - June 2024 - [Devconf.cz](https://devconf.cz) (Marta Anon)
 - August 2024 - [Devconf.us](https://devconf.us) (Sam Koved)
- Applying for admission to CNCF sandbox, very long queue for review
- Already out in the open source community
- Rename on the horizon, still pending on RedHat's legal team



KubeArchive

What does KubeArchive do?

- KubeArchive does three things
 - Archives Kubernetes resources
 - Removes Kubernetes resources
 - Exposes Kubernetes resources
- Components
 - Operator
 - Sink
 - API Server
 - CLI
 - Vacuums



KubeArchive

How does KubeArchive work?

- Two main custom resources
 - KubeArchiveConfig (namespaced)
 - ClusterKubeArchiveConfig (cluster-wide)
- CEL expressions determine what is archived/deleted
 - Allows fine-grained control of what is processed
- Can be configured at namespace level or cluster-wide
 - Cluster configuration takes precedence
- Vacuums to archive and delete older resources
 - Allow processing of resources no longer being updated



KubeArchive

KubeArchive Archiving - archiveWhen

- CEL expression
 - If expression evaluates to true, resource is archived
- Example:

```
---
apiVersion: kubearchive.org/v1
kind: KubeArchiveConfig
metadata:
  name: kubearchive
spec:
  resources:
    - archiveWhen: status.state != 'Completed'
      selector:
        apiVersion: batch/v1
        kind: Job
```



KubeArchive

KubeArchive Deleting - deleteWhen (controller only)

- CEL expression
 - If expression evaluates to true, resource is deleted
- Archives before deleting
- Example:

```
---
apiVersion: kubearchive.org/v1
kind: KubeArchiveConfig
metadata:
  name: kubearchive
spec:
  resources:
    - deleteWhen: status.state = 'Completed'
      selector:
        apiVersion: batch/v1
        kind: Job
```



KubeArchive

KubeArchive Deleting - keepLastWhen (vacuum only)

- CEL expression
 - Keep a certain number of resources matching a CEL expression while deleting the others
- Archives before deleting
- Example:

```
---
apiVersion: kubearchive.org/v1
kind: KubeArchiveConfig
metadata:
  name: kubearchive
spec:
  resources:
    - selector:
        apiVersion: batch/v1
        kind: Job
      keepLastWhen:
        - name: keep-jobs    # Name for this keepLastWhen item
          count: 10
          when: 'keep' in metadata.labels
```



KubeArchive

KubeArchive Bonus - archiveOnDelete (controller only)

- CEL expression
 - If expression evaluates to true, resource is archived
- Archives when delete happens **inside** or **outside** of KubeArchive
- Example:

```
---
apiVersion: kubearchive.org/v1
kind: KubeArchiveConfig
metadata:
  name: kubearchive
spec:
  resources:
    - archiveOnDelete: true
      selector:
        apiVersion: batch/v1
        kind: Job
```



KubeArchive

KubeArchive Vacuums

- Cluster and namespaced custom resources
 - Only one cluster vacuum in the kubearchive namespace
 - Vacuums combine resources defined in both KubeArchiveConfig and ClusterKubeArchiveConfig
- Fill the gap when no messages are sent
 - Delete a PipelineRun two hours after it is completed
- Can be run as Jobs or CronJobs
- Take care of keepLastWhen archival/deletion



What KubeArchive is NOT!

- KubeArchive is not a backup system
 - No restore functionality
- KubeArchive is not a source control system
 - No revisions and no history
- KubeArchive is not a logging system
 - Logging is an integration (not required)
 - Supports anything that accepts a GET request
 - Currently tested with Splunk, Elasticsearch, and Loki (others tested as needed)



KubeArchive

Differences with Tekton Results

- KubeArchive is generic, works for any resource
- KubeArchive does not store logs directly, it relies on logging systems
- KubeArchive uses Common Expression Language for archiving/deleting rules
- The KubeArchive API mimics the Kubernetes API
- The KubeArchive CLI mimics the `get` and `logs` command from `kubectl`
- Read and write operations can be scaled independently



KubeArchive

KubeArchive for Konflux

- Namespace configuration

```
---
apiVersion: kubearchive.org/v1
kind: KubeArchiveConfig
metadata:
  name: kubearchive
spec:
  resources: []
```



KubeArchive for Konflux

- Cluster configuration

```
---
apiVersion: kubearchive.org/v1
kind: ClusterKubeArchiveConfig
metadata:
  name: kubearchive
spec:
  resources:
  - archiveOnDelete: "true"
    selector:
      apiVersion: appstudio.redhat.com/v1alpha1
      kind: Snapshot
  - archiveOnDelete: "true"
    archiveWhen: has(status.completionTime)
    selector:
      apiVersion: appstudio.redhat.com/v1alpha1
      kind: Release
```



KubeArchive

KubeArchive for Konflux

- Cluster configuration (continued)

```
- archiveOnDelete: "true"
  archiveWhen: has(status.completionTime)
  selector:
    apiVersion: tekton.dev/v1
    kind: PipelineRun
- archiveOnDelete: "true"
  archiveWhen: has(status.completionTime)
  selector:
    apiVersion: tekton.dev/v1
    kind: TaskRun
- archiveOnDelete: has(metadata.labels) && "tekton.dev/taskRunUID" in metadata.labels
  archiveWhen: has(metadata.labels) && "tekton.dev/taskRunUID" in metadata.labels
    && status.phase in ['Succeeded', 'Failed', 'Unknown']
  selector:
    apiVersion: v1
    kind: Pod
```



KubeArchive

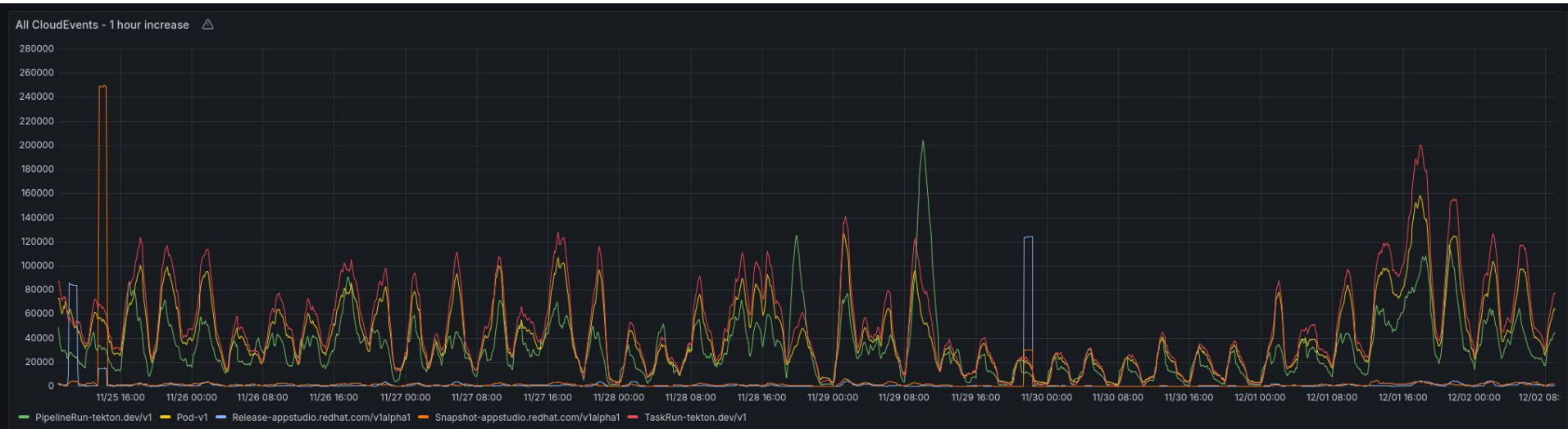
KubeArchive Performance

- Queries designed in consultation with a Postgres expert
 - Performance has been built in from day one
 - Tested with 7.8M Tekton Results dataset
- Installed in an internal Red Hat system



KubeArchive

Some numbers



Received updates per hour

Some numbers



Read latency

Some numbers



Ingestion latency

KubeArchive in the Tekton ecosystem

- Alternative to Tekton Results when other resources need to be stored
- The KubeArchive team wants to donate KubeArchive to the Tekton Community



KubeArchive

Links

- Project: <https://github.com/kubearchive/kubearchive>
- Documentation: <https://kubearchive.github.io/kubearchive>
- Email lists: <https://groups.google.com/g/kubearchive>,
<https://groups.google.com/g/kubearchive-developers>
- Slack: <https://app.slack.com/client/T09NY5SBT/C07MB5YBVCL>

Follow us on GitHub!



KubeArchive